

Furuta Pendulum Sim-to-Real DR Report

June 4, 2026 - No DR vs Stage 2 Domain Randomization

Summary: The clean no-DR policy solves the nominal MuJoCo task, but it is fragile under realistic sim-to-real perturbations. Stage 2 DR adds small elbow friction, sensor noise, motor variation, delay, and encoder zero offsets. It lowers perfect-sim elegance but strongly improves worst-case robustness.

Current Model

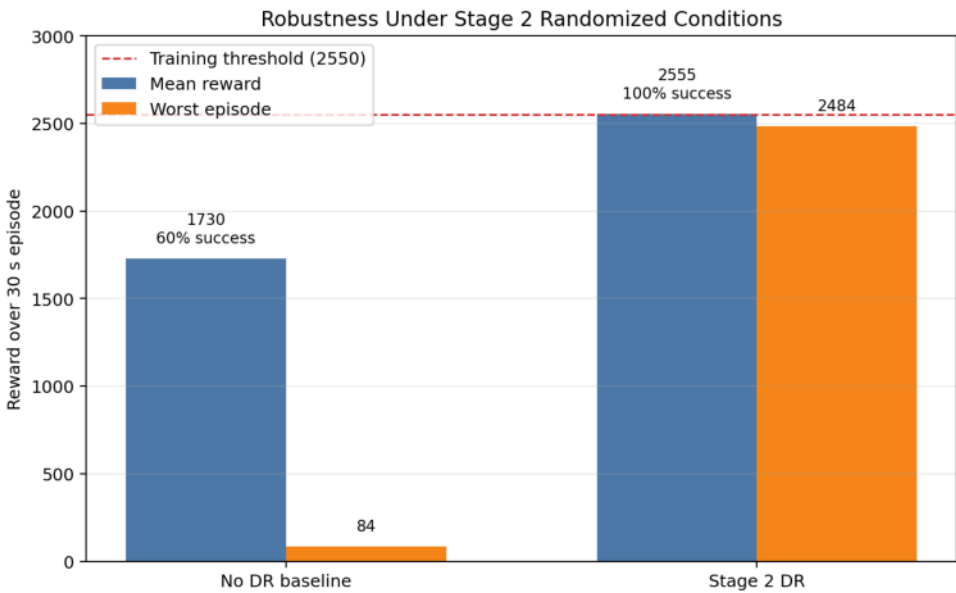
- ? CAD: arm_2.STL, rod_2.STL
- ? Elbow convention: qpos[1] = 0 upright, qpos[1] = +/-pi hanging
- ? Rod mass: 0.01888 kg; pivot-to-COM: 0.03603 m
- ? Control period: 10 ms; episode length: 30 s; maximum reward: 3000

Stage 2 DR Profile

mass/inertia +/-5% shoulder damping +/-5% elbow damping 0 to 0.00002 N*m*s/rad motor +/-5%, action delay 0-1 step, shoulder/elbow velocity filtering is excluded.

Robustness:

Both policies we



Policy	Mean	Std	Worst	Best	Success
No DR baseline	1729.64	1336.33	84.20	2893.91	60%
Stage 2 DR	2554.67	43.38	2483.71	2638.26	100%

Policy	Swing-up	Action RMS	Control Energy	Shoulder Range
No DR baseline	0.96 s	0.347	5.07	136.1 deg
Stage 2 DR	1.02 s	0.372	4.18	271.7 deg

Key finding: Stage 2 improves worst-case reward from 84.20 to 2483.71 and success rate from 60% to 100%.

Next: disturbance recovery tests, elbow free-swing friction measurement, AS5600 velocity-lag measurement, then board-tilt DR.